- **Rear Axle Lock Request Input** - Each time the cluster detects ground on the rear axle lock request circuit from the axle lock switch (rear axle lock request circuit ground = rear axle lock request active) while the key is in the ignition lock cylinder, the rear lock indicator flashes on and off at a slow rate. The indicator continues to flash until the rear axle lock request input to the cluster is an open circuit (rear axle lock request circuit open = rear axle lock request inactive), or until the cluster detects a change in the inputs from the rear locker indicator switch or the key-in ignition switch, whichever occurs first.
- **Rear Locker Indicator Switch Input** - Each time the cluster detects ground on the rear locker indicator switch sense circuit (rear locker indicator switch closed = rear axle locker engaged) while the rear axle lock request is active and the key is in the ignition lock cylinder, the rear lock indicator is illuminated solid. The indicator remains illuminated solid until the rear locker indicator switch input to the cluster is an open circuit (rear locker indicator switch open = rear axle locker disengaged), or until the cluster detects a change in the inputs from the axle lock switch or the key-in ignition switch, whichever occurs first.
- **Rear Lock Request/Feedback Mismatch** - Each time the cluster detects that the inputs from the axle lock switch (request) on the rear axle lock request circuit and from the rear locker indicator switch (feedback) on the rear locker indicator switch sense circuit are mismatched, the rear lock indicator flashes on and off at a slow rate. This condition occurs when the lock request is active but the axle locker remains disengaged, or when the lock request is inactive but the axle locker remains engaged. The indicator continues to flash until the request and the feedback are matching.
- **Key-In Ignition Switch Input** - Each time the cluster detects that the key has been removed from the ignition lock cylinder (key-in ignition switch open = key is not in ignition switch) while the rear axle lock request is still active, the rear lock indicator flashes at a fast rate and three chime tones are sounded. The indicator will continue to flash at this rate until the key is replaced in the ignition switch lock cylinder (key-in ignition switch closed = key is in ignition switch) while the rear axle lock request is still active, or until the rear axle lock request input is deactivated, whichever occurs first.

The axle lock switch in the accessory switch bezel on the instrument panel provides a hard wired ground input to the instrument cluster circuitry through the rear axle lock request circuit whenever the switch is enabled and the rear or the rear and front axle lock position of the switch is selected by the vehicle operator. The rear locker indicator switch on the rear axle housing provides a hard wired ground input to the instrument cluster whenever the rear axle locker mechanism is engaged. The key-in ignition switch integral to the ignition switch provides a hard wired ground input to the instrument cluster whenever a key is present in the ignition lock cylinder. Each of these inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the instrument cluster circuitry that controls the rear lock indicator LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

SEATBELT INDICATOR

DESCRIPTION

Fig. 22 Seatbelt Indicator

A seatbelt indicator is standard equipment on all instrument clusters (Fig. 22). The seatbelt indicator is located near the center of the instrument cluster, between the tachometer and the speedometer. The seatbelt indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Seat Belt” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The seatbelt indicator is serviced as a unit with the instrument cluster.

OPERATION

The seatbelt indicator gives an indication to the vehicle operator of the status of the driver side front seatbelt. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and a hard wired input from the seatbelt switch in the driver side front seatbelt buckle through the seat belt indicator driver circuit. The seatbelt indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will